Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: David Machado (former F&S Brasil Dairy)

Physical address of dairy:

Number and Street	City	County	Zip Code
26230 S Union RD	Manteca	San Joaquin	95337

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X257-XX12-X001-XXXX X257-XX12-X002-XXXX

B. OPERATORS

Machado, David

Operator name: Machado, David

Telephone no.: Landline: ; Cellular: (209) 581-1713

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1046	Ripon	CA	95366

This operator is responsible for paying permit fees.

C. OWNERS

Machado, David

Legal owner name: Machado, David

Telephone no.: Landline: ; Cellular: (209) 581-1713

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1046	Ripon	CA	95366

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	360	0	0	0	0	0
Maximum number	1,440	0	0	0	0	0
Average number	360	0	0	0	0	0
Avg live weight (lbs)	1,200	0	0	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 9,149.52 *tons per reporting period*

Total nitrogen from manure: 119,632.02 *lbs per reporting period*

After ammonia losses (30% loss applied): 83,742.41 *lbs per reporting period*

Total phosphorus from manure: 20,236.88 *lbs per reporting period*

Total potassium from manure: 68,470.28 *lbs per reporting period*

Total salt from manure: 169,506.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 23,724,000 *gallons*

Total nitrogen generated: 53,103.59 *lbs*

Total phosphorus generated: 9,892.62 *lbs*

Total potassium generated: 80,803.07 *lbs*

Total salt generated: 585,022.89 *lbs*

	13,680,000	*gallons applied*
+	10,044,000	*gallons exported*
-	0	*gallons imported*
=	23,724,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW	Ground water
Stanislaus Water Users	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/09/2023	Solid commercial fertilizer *UN 32*	33.50 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
04/23/2023	Solid commercial fertilizer *15-15-15*	3.24 *ton*	Dry-weight	0.1	15.000000	15.000000	15.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	22,389.59	971.03	971.03	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	22,389.59	971.03	971.03	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/02/2023	Corral solids	4,310.00 *ton*	Dry-weight	30.1		12,500.00	20,500.00	4,700.00		49.10
10/30/2023	Corral solids	7,400.00 *ton*	Dry-weight	30.1		12,500.00	20,500.00	4,700.00		49.10

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
03/25/2023	Process wastewater	3,024,000.00 *gal*	252.00	129.00	0.00	0.00	88.30	314.00		2,380
05/24/2023	Process wastewater	2,412,000.00 *gal*	459.00	316.00	0.00	0.00	50.30	588.00		4,660
07/17/2023	Process wastewater	2,304,000.00 *gal*	112.00	53.20	0.00	0.00	26.70	270.00		1,920
08/10/2023	Process wastewater	2,304,000.00 *gal*	112.00	53.20	0.00	0.00	26.70	270.00		1,920

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	204,632.25	335,596.89	76,941.73	8,037,954.78
Process wastewater	19,904.93	4,267.44	30,141.74	227,688.32
Total exports for all materials	224,537.18	339,864.33	107,083.47	8,265,643.10

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	70	70	2	both	X057-XX12-X001-XXXX
2	8	8	2	both	X057-XX12-X001-XXXX
3	47	47	1	none	X057-XX12-X001-XXXX
4	47	47	1	none	X057-XX12-X001-XXXX
5	12	12	1	none	X057-XX12-X001-XXXX
6	14	14	2	both	X257-XX12-X002-XXXX
7	54	54	1	none	X257-XX12-X002-XXXX
Manteca Road Almonds	103	103	1	none	X257-XX17-X039-XXXX X257-XX17-X040-XXXX X257-XX17-X041-XXXX X257-XX17-X042-XXXX X257-XX17-X044-XXXX X257-XX17-X045-XXXX X257-XX17-X046-XXXX X257-XX17-X048-XXXX X257-XX17-X058-XXXX X257-XX17-X059-XXXX X257-XX17-X060-XXXX
Totals for areas that were used for application	92	92	6		
Totals for areas that were not used for application	263	263	5		
Land application area totals	355	355	11		

B. CROPS AND HARVESTS

1

Field name: 1

1

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 70 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	784.00 *ton*	As-is		57.4	9,030.00	1,600.00	12,600.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.20	202.27	35.84	282.24	1,049.66

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,887.00 *ton*	As-is		69.4	3,830.00	740.00	5,910.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.96	206.49	39.90	318.63	1,138.35

2

Field name: 2

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	94.00 *ton*	As-is		60.6	8,310.00	1,490.00	12,700.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.75	195.29	35.02	298.45	1,129.60

2

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	184.00 *ton*	As-is		66.2	4,330.00	850.00	5,570.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.00	199.18	39.10	256.22	730.76

3

Field name: 3

10/15/2018: Almond Field Run

Crop: Almond Field Run Acres planted: 47 Plant date: 10/15/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	191.00 *ton*	As-is		11.4	19,800.00	2,290.00	31,700.00		22.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.06	160.93	18.61	257.65	1,605.85

4

Field name: 4

4

03/16/2017: Almond Field Run

Crop: Almond Field Run Acres planted: 47 Plant date: 03/16/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	206.00 *ton*	As-is		11.4	19,800.00	2,290.00	31,700.00		22.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.38	173.57	20.07	277.88	1,731.96

5

Field name: 5

03/16/2017: Almond Field Run

Crop: Almond Field Run Acres planted: 12 Plant date: 03/16/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	51.00 *ton*	As-is		11.4	19,800.00	2,290.00	31,700.00		22.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.25	168.30	19.47	269.45	1,679.41

6

Field name: 6

6

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	234.00 *ton*	As-is		72.4	4,660.00	910.00	7,660.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.71	155.78	30.42	256.06	977.99

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	338.00 *ton*	As-is		65.8	4,210.00	790.00	5,940.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.14	203.28	38.15	286.82	1,056.88

7

Field name: 7

01/24/2022: Almond Field Run

Crop: Almond Field Run Acres planted: 54 Plant date: 01/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.01 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Manteca Road Almonds

Field name: Manteca Road Almonds

02/03/2015: Almond Field Run

Crop: Almond Field Run Acres planted: 97 Plant date: 02/03/2015

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	399.00 *ton*	As-is		11.4	19,800.00	2,300.00	31,700.00		22.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.11	162.89	18.92	260.79	1,625.44

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/26/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.88	10.37	135.21	721.35	1,584,000.00 *gal*
Stanislaus Water Users	Ground water	3.97	0.00	0.00	0.00	11,880,000.00 *gal*
Application event totals		83.84	10.37	135.21	721.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.89	22.74	80.86	612.86	2,160,000.00 *gal*
IW	Ground water	20.60	0.00	0.00	319.73	3,600,000.00 *gal*
Application event totals		85.49	22.74	80.86	932.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.80	8.63	100.94	799.98	1,440,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	46.35	10,800,000.00 *gal*
Application event totals		78.80	8.63	100.94	846.33	

1 - 06/25/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	131.80	21.01	25.54	1,081.20	1,360.00 *ton*
Application event totals		131.80	21.01	25.54	1,081.20	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.62	9.93	116.08	919.97	1,656,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	53.30	12,420,000.00 *gal*
Application event totals		90.62	9.93	116.08	973.27	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	46.35	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.35	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.23	4.58	46.35	329.60	1,440,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	46.35	10,800,000.00 *gal*
Application event totals		19.23	4.58	46.35	375.95	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.30	4.13	41.72	296.64	1,296,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	41.72	9,720,000.00 *gal*
Application event totals		17.30	4.13	41.72	338.36	

1 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	39.40	9,180,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.38	3.67	37.08	263.68	1,152,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	37.08	8,640,000.00 *gal*
Application event totals		15.38	3.67	37.08	300.76	

2 - 11/26/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.42	10.31	134.44	717.25	180,000.00 *gal*
Stanislaus Water Users	Ground water	3.94	0.00	0.00	0.00	1,350,000.00 *gal*
Application event totals		83.37	10.31	134.44	717.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.71	26.53	94.33	715.00	288,000.00 *gal*
IW	Ground water	24.03	0.00	0.00	373.02	480,000.00 *gal*
Application event totals		99.74	26.53	94.33	1,088.02	

2 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.95	7.56	88.32	699.98	144,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	40.56	1,080,000.00 *gal*
Application event totals		68.95	7.56	88.32	740.54	

2 - 06/25/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	127.20	20.27	24.65	1,043.44	150.00 *ton*
Application event totals		127.20	20.27	24.65	1,043.44	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.80	10.39	121.44	962.47	198,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	55.77	1,485,000.00 *gal*
Application event totals		94.80	10.39	121.44	1,018.24	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	40.56	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.56	

2 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.82	4.01	40.56	288.40	144,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	40.56	1,080,000.00 *gal*
Application event totals		16.82	4.01	40.56	328.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.72	3.51	35.49	252.35	126,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	35.49	945,000.00 *gal*
Application event totals		14.72	3.51	35.49	287.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	30.42	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	12.62	3.01	30.42	216.30	108,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	30.42	810,000.00 *gal*
Application event totals		12.62	3.01	30.42	246.72	

3 - 10/15/2018: Almond Field Run

Field name: 3

Crop: Almond Field Run

Plant date: 10/15/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 10/15/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	1.98	0.00	0.00	0.00	3,981,699.00 *gal*
Application event totals		1.98	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.90	7,963,398.00 *gal*
Application event totals		0.00	0.00	0.00	50.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.90	7,963,398.00 *gal*
Application event totals		0.00	0.00	0.00	50.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.90	7,963,398.00 *gal*
Application event totals		0.00	0.00	0.00	50.90	

3 - 10/15/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.90	7,963,398.00 *gal*
Application event totals		0.00	0.00	0.00	50.90	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.90	7,963,398.00 *gal*
Application event totals		0.00	0.00	0.00	50.90	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	25.45	3,981,699.00 *gal*
Application event totals		0.00	0.00	0.00	25.45	

4 - 03/16/2017: Almond Field Run

Field name: 4

Crop: Almond Field Run

Plant date: 03/16/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	27.62	4,321,482.00 *gal*
Application event totals		0.00	0.00	0.00	27.62	

4 - 03/16/2017: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	55.25	8,642,964.00 *gal*
Application event totals		0.00	0.00	0.00	55.25	

04/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	55.25	8,642,964.00 *gal*
Application event totals		0.00	0.00	0.00	55.25	

05/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	55.25	8,642,964.00 *gal*
Application event totals		0.00	0.00	0.00	55.25	

06/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

4 - 03/16/2017: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	55.25	8,642,964.00 *gal*
Application event totals		0.00	0.00	0.00	55.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	55.25	8,642,964.00 *gal*
Application event totals		0.00	0.00	0.00	55.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	27.62	4,321,482.00 *gal*
Application event totals		0.00	0.00	0.00	27.62	

5 - 03/16/2017: Almond Field Run

Field name: 5

Crop: Almond Field Run

Plant date: 03/16/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	25.57	1,021,526.00 *gal*
Application event totals		0.00	0.00	0.00	25.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	51.15	2,043,052.00 *gal*
Application event totals		0.00	0.00	0.00	51.15	

5 - 03/16/2017: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	51.15	2,043,052.00 *gal*
Application event totals		0.00	0.00	0.00	51.15	

05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	51.15	2,043,052.00 *gal*
Application event totals		0.00	0.00	0.00	51.15	

06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	51.15	2,043,052.00 *gal*
Application event totals		0.00	0.00	0.00	51.15	

5 - 03/16/2017: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	51.15	2,043,052.00 *gal*
Application event totals		0.00	0.00	0.00	51.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	25.57	1,021,526.00 *gal*
Application event totals		0.00	0.00	0.00	25.57	

6 - 11/26/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.77	11.78	153.64	819.72	360,000.00 *gal*
Stanislaus Water Users	Ground water	4.51	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		95.28	11.78	153.64	819.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.49	10.79	126.18	999.97	360,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	57.94	2,700,000.00 *gal*
Application event totals		98.49	10.79	126.18	1,057.91	

6 - 06/25/2023: Corn, silage

6 - 06/25/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	130.83	20.85	25.35	1,073.25	270.00 *ton*
Application event totals		130.83	20.85	25.35	1,073.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.49	10.79	126.18	999.97	360,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	57.94	2,700,000.00 *gal*
Application event totals		98.49	10.79	126.18	1,057.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	46.35	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.82	4.01	40.56	288.40	252,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	40.56	1,890,000.00 *gal*
Application event totals		16.82	4.01	40.56	328.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.42	3.44	34.76	247.20	216,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	34.76	1,620,000.00 *gal*
Application event totals		14.42	3.44	34.76	281.97	

6 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	40.56	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.42	3.44	34.76	247.20	216,000.00 *gal*
Stanislaus Water Users	Ground water	0.00	0.00	0.00	34.76	1,620,000.00 *gal*
Application event totals		14.42	3.44	34.76	281.97	

7 - 01/24/2022: Almond Field Run

Field name: 7

Crop: Almond Field Run

Plant date: 01/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.98	9,163,992.00 *gal*
Application event totals		0.00	0.00	0.00	50.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
15-15-15	Solid commercial fertilizer	6.00	6.00	6.00	0.00	
Application event totals		6.00	6.00	6.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.98	9,163,992.00 *gal*
Application event totals		0.00	0.00	0.00	50.98	

7 - 01/24/2022: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.98	9,163,992.00 *gal*
Application event totals		0.00	0.00	0.00	50.98	

06/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
15-15-15	Solid commercial fertilizer	6.00	6.00	6.00	0.00	
Application event totals		6.00	6.00	6.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.98	9,163,992.00 *gal*
Application event totals		0.00	0.00	0.00	50.98	

07/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
15-15-15	Solid commercial fertilizer	6.00	6.00	6.00	0.00	
Application event totals		6.00	6.00	6.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	50.98	9,163,992.00 *gal*
Application event totals		0.00	0.00	0.00	50.98	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	25.49	4,581,996.00 *gal*
Application event totals		0.00	0.00	0.00	25.49	

Manteca Road Almonds - 02/03/2015: Almond Field Run

Manteca Road Almonds - 02/03/2015: Almond Field Run

Field name: Manteca Road Almonds

Crop: Almond Field Run

Plant date: 02/03/2015

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	2.30	0.00	0.00	0.00	9,565,929.00 *gal*
Application event totals		2.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	59.25	19,131,858.00 *gal*
Application event totals		0.00	0.00	0.00	59.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	59.25	19,131,858.00 *gal*
Application event totals		0.00	0.00	0.00	59.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	59.25	19,131,858.00 *gal*
Application event totals		0.00	0.00	0.00	59.25	

Manteca Road Almonds - 02/03/2015: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	59.25	19,131,858.00 *gal*
Application event totals		0.00	0.00	0.00	59.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	59.25	19,131,858.00 *gal*
Application event totals		0.00	0.00	0.00	59.25	

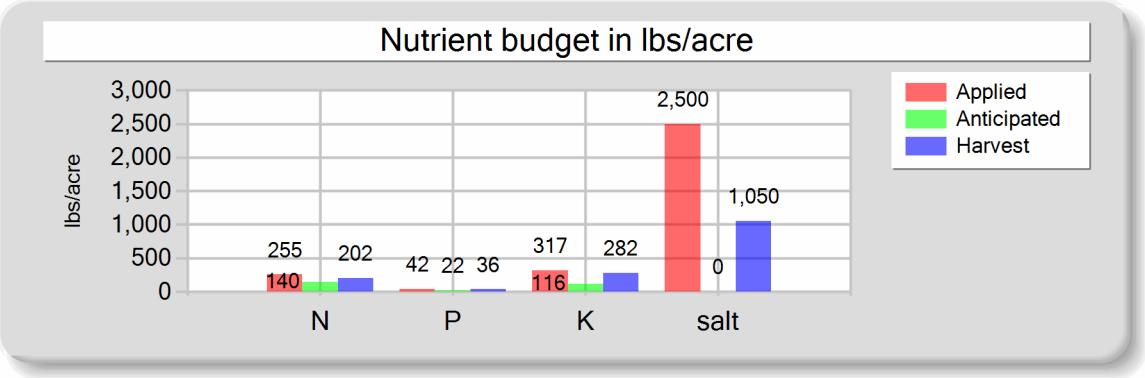
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users	Ground water	0.00	0.00	0.00	29.63	9,565,929.00 *gal*
Application event totals		0.00	0.00	0.00	29.63	

B. NUTRIENT BUDGET

1 - 11/26/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.56	41.74	317.00	2,134.18
Fresh water	24.57	0.00	0.00	366.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.13	41.74	317.00	2,500.27
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	202.27	35.84	282.24	1,049.66
Nutrient balance	52.86	5.90	34.76	1,450.60
Applied to removed ratio	1.26	1.16	1.12	2.38

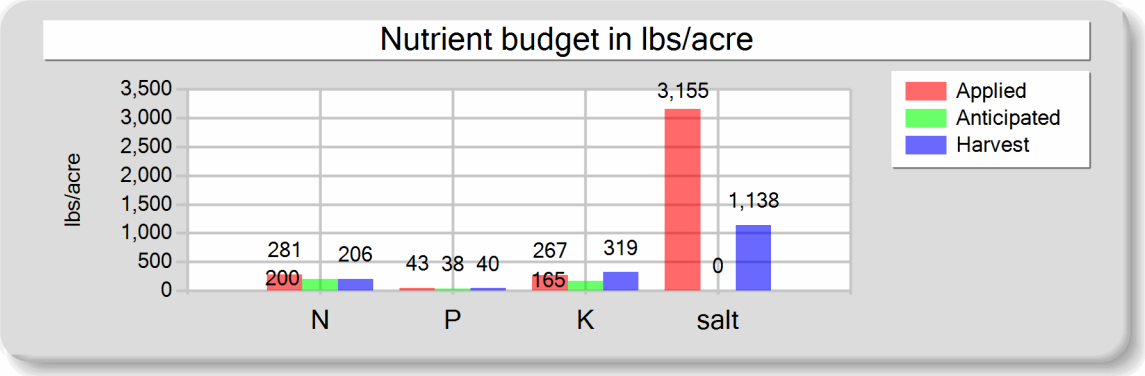
Fresh water applied	
26,280,000.00	*gallons*
967.80	*acre-inches*
13.83	*inches/acre*

Process wastewater applied	
5,184,000.00	*gallons*
190.91	*acre-inches*
2.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/25/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	131.80	21.01	25.54	1,081.20
Process wastewater	142.53	22.31	241.23	1,809.90
Fresh water	0.00	0.00	0.00	264.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.33	43.31	266.77	3,155.30
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	206.49	39.90	318.63	1,138.35
Nutrient balance	74.84	3.42	-51.87	2,016.95
Applied to removed ratio	1.36	1.09	0.84	2.77

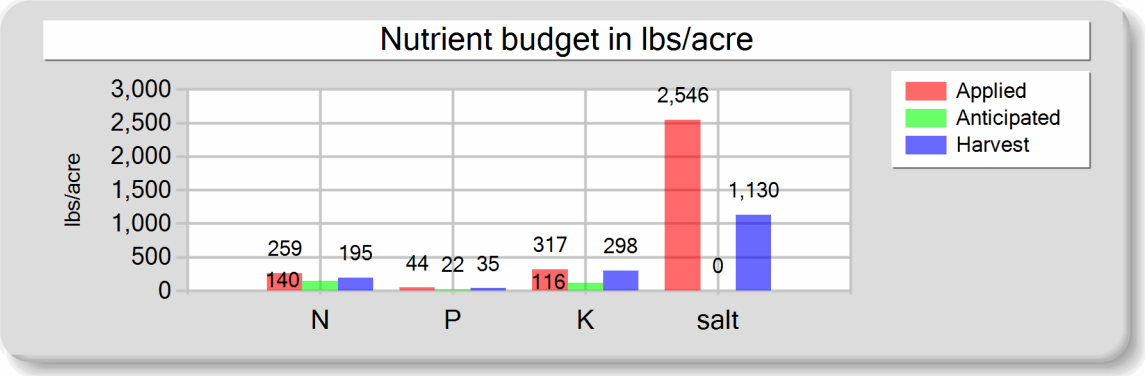
Fresh water applied	
61,560,000.00	*gallons*
2,267.05	*acre-inches*
32.39	*inches/acre*

Process wastewater applied	
5,544,000.00	*gallons*
204.17	*acre-inches*
2.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/26/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.08	44.39	317.09	2,132.23
Fresh water	27.98	0.00	0.00	413.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.05	44.39	317.09	2,545.81
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	195.29	35.02	298.45	1,129.60
Nutrient balance	63.77	9.38	18.64	1,416.21
Applied to removed ratio	1.33	1.27	1.06	2.25

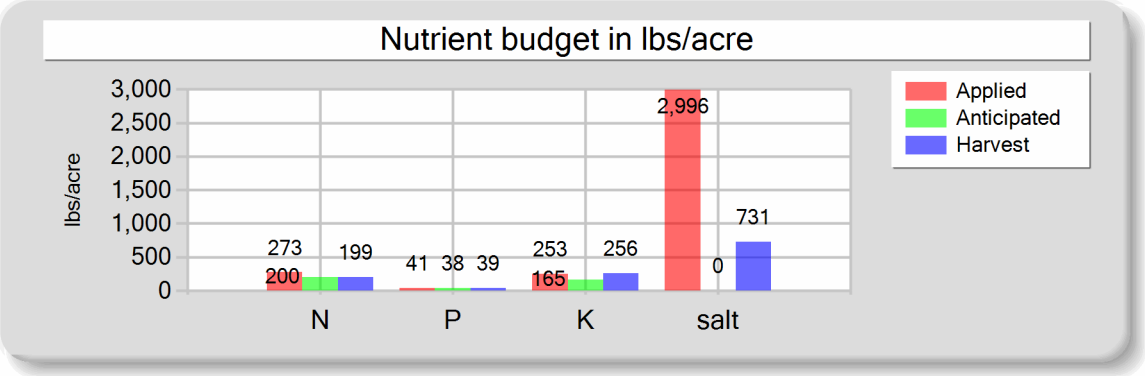
Fresh water applied	
2,910,000.00	*gallons*
107.17	*acre-inches*
13.40	*inches/acre*

Process wastewater applied	
612,000.00	*gallons*
22.54	*acre-inches*
2.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/25/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.20	20.27	24.65	1,043.44
Process wastewater	138.96	20.92	227.91	1,719.53
Fresh water	0.00	0.00	0.00	233.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	273.16	41.19	252.55	2,996.17
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	199.18	39.10	256.22	730.76
Nutrient balance	73.98	2.09	-3.67	2,265.41
Applied to removed ratio	1.37	1.05	0.99	4.10

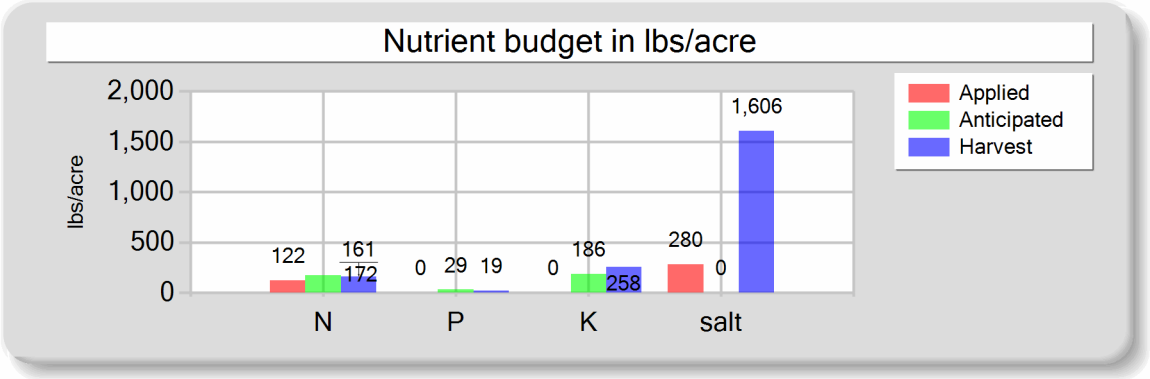
Fresh water applied	
6,210,000.00	*gallons*
228.69	*acre-inches*
28.59	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
2.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/15/2018: Almond Field Run

Field name: 3 Crop: Almond Field Run Plant date: 10/15/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.98	0.00	0.00	279.96
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	121.58	0.00	0.00	279.96
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	160.93	18.61	257.65	1,605.85
Nutrient balance	-39.35	-18.61	-257.65	-1,325.89
Applied to removed ratio	0.76	0.00	0.00	0.17

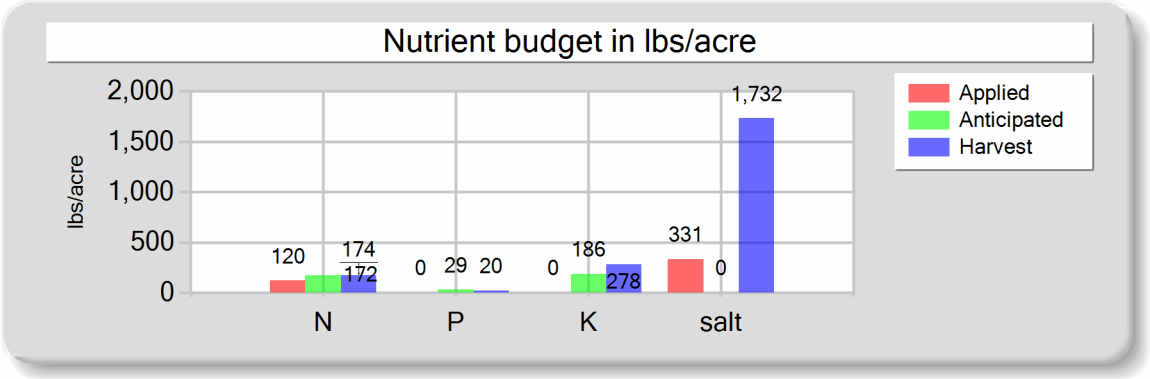
Fresh water applied	
47,780,388.00	*gallons*
1,759.59	*acre-inches*
37.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 03/16/2017: Almond Field Run

Field name: 4 Crop: Almond Field Run Plant date: 03/16/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	331.47
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	119.60	0.00	0.00	331.47
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	173.57	20.07	277.88	1,731.96
Nutrient balance	-53.97	-20.07	-277.88	-1,400.49
Applied to removed ratio	0.69	0.00	0.00	0.19

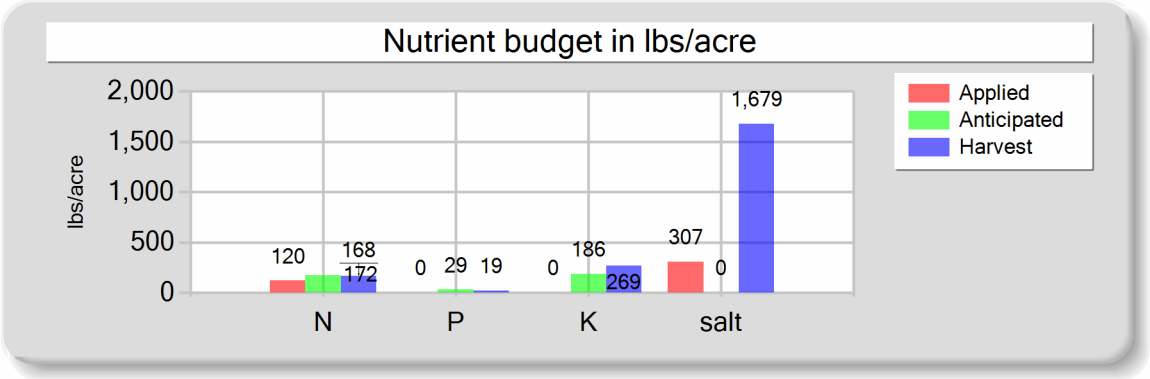
Fresh water applied	
51,857,784.00	*gallons*
1,909.75	*acre-inches*
40.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 03/16/2017: Almond Field Run

Field name: 5 Crop: Almond Field Run Plant date: 03/16/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	306.89
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	119.60	0.00	0.00	306.89
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	168.30	19.47	269.45	1,679.41
Nutrient balance	-48.70	-19.47	-269.45	-1,372.53
Applied to removed ratio	0.71	0.00	0.00	0.18

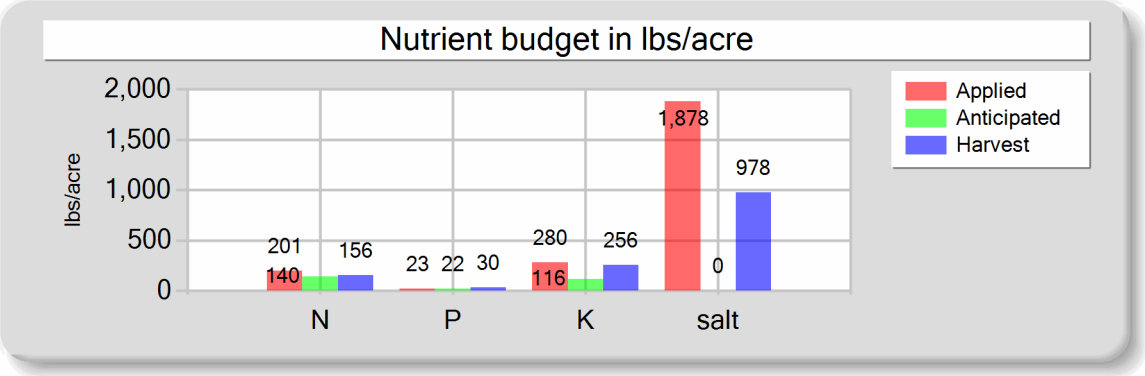
Fresh water applied	
12,258,312.00	*gallons*
451.43	*acre-inches*
37.62	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/26/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.26	22.57	279.82	1,819.69
Fresh water	4.51	0.00	0.00	57.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	200.77	22.57	279.82	1,877.63
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	155.78	30.42	256.06	977.99
Nutrient balance	44.99	-7.85	23.76	899.64
Applied to removed ratio	1.29	0.74	1.09	1.92

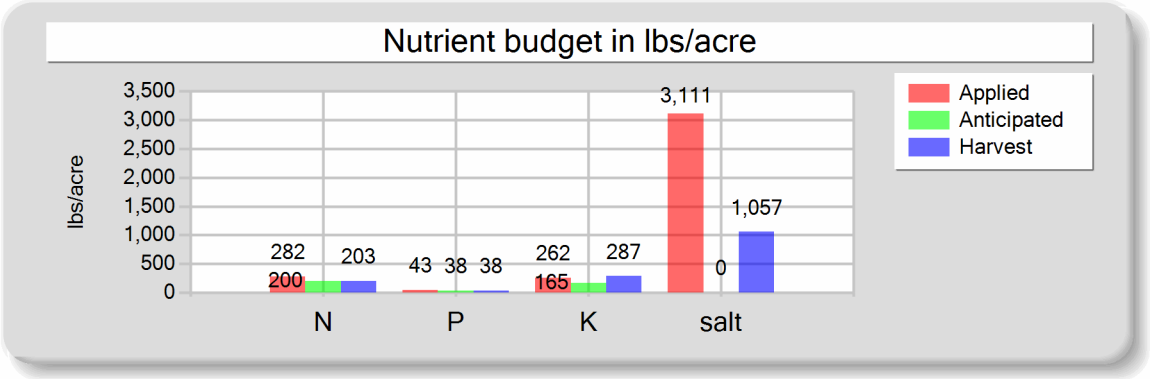
Fresh water applied	
5,400,000.00	*gallons*
198.86	*acre-inches*
14.20	*inches/acre*

Process wastewater applied	
720,000.00	*gallons*
26.52	*acre-inches*
1.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/25/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	130.83	20.85	25.35	1,073.25
Process wastewater	144.16	21.68	236.26	1,782.78
Fresh water	0.00	0.00	0.00	254.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.99	42.53	261.61	3,110.96
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	203.28	38.15	286.82	1,056.88
Nutrient balance	78.71	4.39	-25.21	2,054.08
Applied to removed ratio	1.39	1.11	0.91	2.94

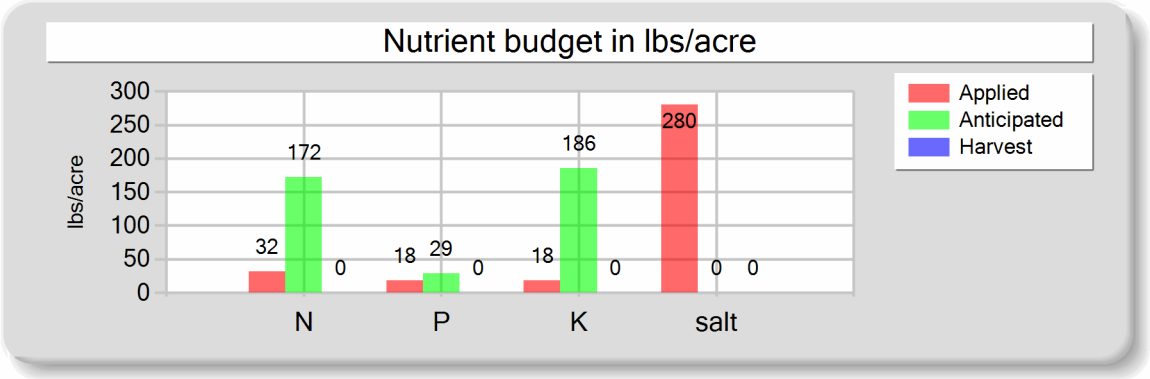
Fresh water applied	
11,880,000.00	*gallons*
437.50	*acre-inches*
31.25	*inches/acre*

Process wastewater applied	
1,044,000.00	*gallons*
38.45	*acre-inches*
2.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 01/24/2022: Almond Field Run

Field name: 7 Crop: Almond Field Run Plant date: 01/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18.00	18.00	18.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	280.40
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	32.00	18.00	18.00	280.40
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	32.00	18.00	18.00	280.40
Applied to removed ratio	0.00	0.00	0.00	0.00

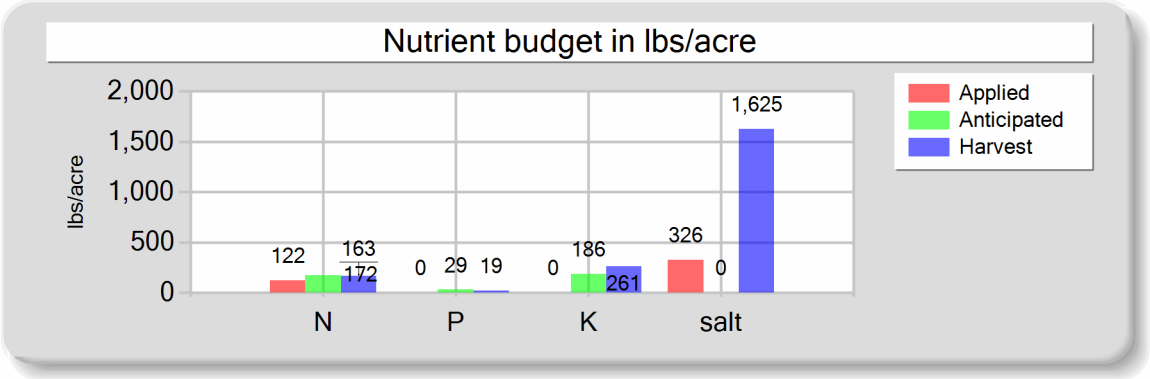
Fresh water applied	
50,401,956.00	*gallons*
1,856.13	*acre-inches*
34.37	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manteca Road Almonds - 02/03/2015: Almond Field Run

Field name: Manteca Road Almonds Crop: Almond Field Run Plant date: 02/03/2015



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.30	0.00	0.00	325.89
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	121.90	0.00	0.00	325.89
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	162.89	18.92	260.79	1,625.44
Nutrient balance	-40.99	-18.92	-260.79	-1,299.54
Applied to removed ratio	0.75	0.00	0.00	0.20

Fresh water applied	
114,791,148.00	*gallons*
4,227.37	*acre-inches*
43.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 31.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	9,470.00	14,700.00							58.20
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/28/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	2,040.00	2,480.00							10.50
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 11/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 30.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	20,500.00	4,700.00							49.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	423.00	238.00	0.00	0.00	54.90	716.00								6,050.00	3,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 01/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	252.00	129.00	0.00	0.00	88.30	314.00								3,060.00	2,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	459.00	316.00	0.00	0.00	50.30	588.00	160.00	103.00	179.00	2,860.00	0.00	303.00	307.00	5,360.00	4,660
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	50.00	1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 08/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	112.00	53.20	0.00	0.00	26.70	270.00								2,440.00	1,920
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 11/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	106.00	56.00	0.00	0.00	37.20	198.00								15,800.00	1,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW

Fresh Water

Sample description: Fresh Water

Sample date: 03/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	48.00	0.00	45.20								1,370.00	745
DL	0.50	0.50	0.50								1.00	10

Stanislaus Water Users

Fresh Water

Sample description: Fresh Water

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								60.00	36
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,030.00	1,600.00	12,600.00		11.00
DL	100.00	100.00	100.00		0.01

1 - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,260.00	680.00	5,770.00		7.00
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,830.00	740.00	5,910.00		6.90
DL	100.00	100.00	100.00		0.01

2 - 11/26/2022: Oats, silage-soft dough

2 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,310.00	1,490.00	12,700.00		12.20
DL	100.00	100.00	100.00		0.01

2 - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,330.00	850.00	5,570.00		4.70
DL	100.00	100.00	100.00		0.01

3 - 10/15/2018: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	2,290.00	31,700.00		22.30
DL	100.00	100.00	100.00		0.01

4 - 03/16/2017: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	2,290.00	31,700.00		22.30
DL	100.00	100.00	100.00		0.01

5 - 03/16/2017: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	2,290.00	31,700.00		22.30
DL	100.00	100.00	100.00		0.01

6 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,660.00	910.00	7,660.00		10.60
DL	100.00	100.00	100.00		0.01

6 - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,210.00	790.00	5,940.00		6.40
DL	100.00	100.00	100.00		0.01

7 - 01/24/2022: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/07/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	2,290.00	31,700.00		22.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

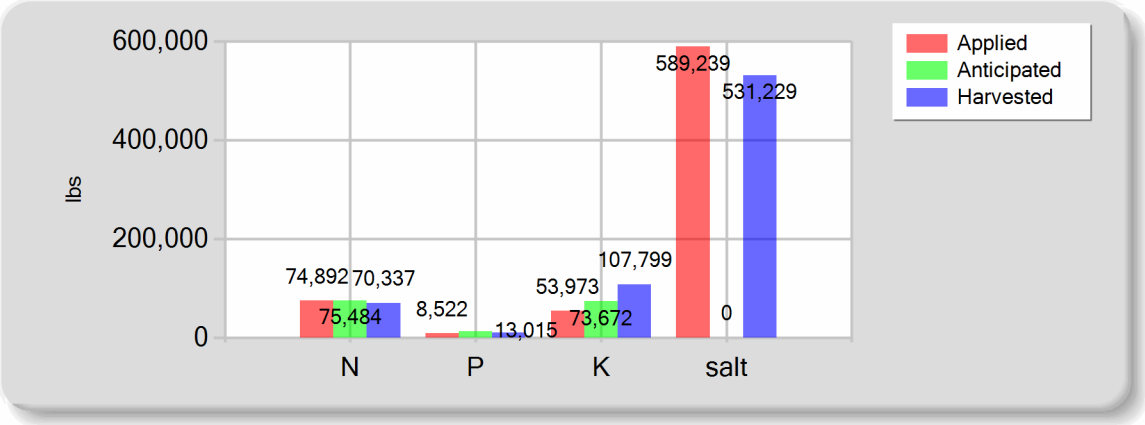
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

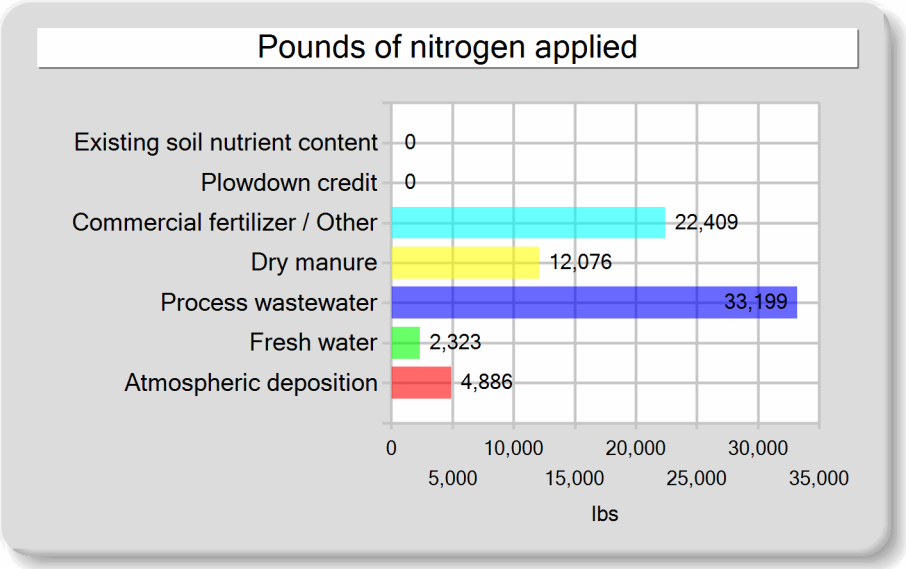
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

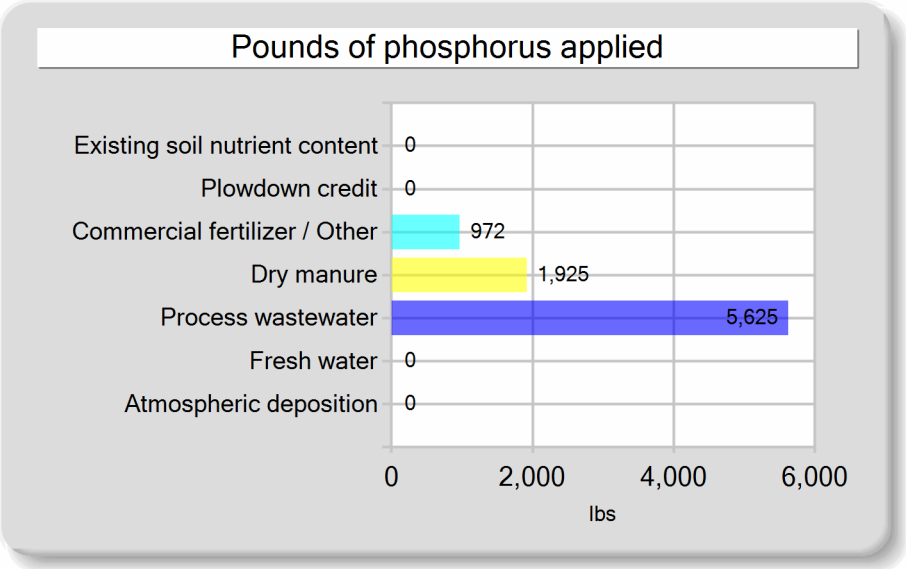
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	22,408.80	972.00	972.00	0.00
Dry manure	12,075.52	1,924.54	2,339.63	99,057.00
Process wastewater	33,198.66	5,625.18	50,661.33	357,334.57
Fresh water	2,323.06	0.00	0.00	132,847.31
Atmospheric deposition	4,886.00	0.00	0.00	0.00
Total nutrients applied	74,892.04	8,521.72	53,972.96	589,238.88
Anticipated crop nutrient removal	75,484.00	13,015.20	73,672.40	0.00
Actual crop nutrient removal	70,337.22	10,741.64	107,798.62	531,228.78
Nutrient balance	4,554.82	-2,219.92	-53,825.66	58,010.11
Applied to removed ratio	1.06	0.79	0.50	1.11

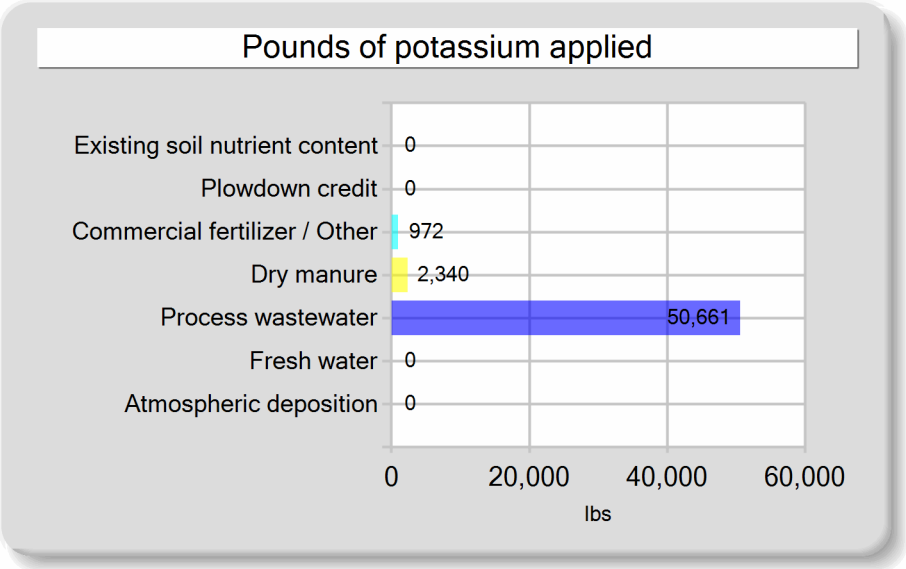
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

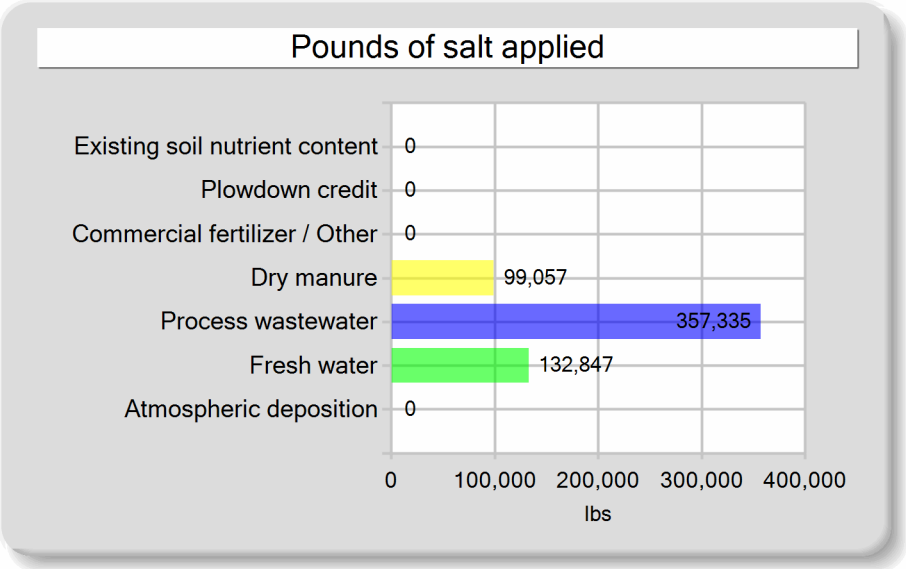


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

All animals were removed from the facility in April 2023. The facility has started the closure process with CVRWQCB.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
David Machado	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: DAVID MACHADO

Name of Dairy Facility: DAVID MACHADO

Facility Address: 26230 S. UNION RD. MANTECA 95337
Number and Street / City / Zip Code

Contact Person Name and Phone Number: DAVID MACHADO 209-581-1713
Name / Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: CAMBRA FARMS

Address of Hauling Company /Person: P.O. BOX 4034 MANTECA 95337
Number and Street / City / Zip Code

Contact Person: BILL CAMBRA 209-823-5363
Name / Phone Number

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CARDOZA BROS 2444 S. AIRPORT WAY MANTECA 95337 209-607-4743
Name / Number and Street / City / Zip Code / Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

2444 S. AIRPORT WAY MANTECA 95337
Number and Street / City / Zip Code / Assessor's Parcel Number

Dates Hauled: 7/2/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 4310 (Tons) or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 30.1% MOISTURE
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

HAULER CALCULATIONS

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes ____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT) Date: Apr 1, 2024

Hauler's Signature: _____ Date: 7/2/23

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: DAVID MACHADO

Name of Dairy Facility: DAVID MACHADO

Facility Address: 26230 S. UNION RD. MANTECA 95337
Number and Street City Zip Code

Contact Person Name and Phone Number: DAVID MACHADO 209-581-1713
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: CAMBRA FARMS

Address of Hauling Company /Person: P.O. BOX 4034 MANTECA 95337
Number and Street City Zip Code

Contact Person: BILL CAMBRA 209-823-5363
Name Phone Number

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CARDOZA BROS 2444 S. AIRPORT WAY MANTECA 95337 209-607-4743
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

2444 S. AIRPORT WAY MANTECA 95337
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 10/30/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 7400 (Tons) or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 30.1% MOISTURE
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

HAULER CALCULATIONS

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

_____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

Hauler's Signature: _____
Date: 10/30/23

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: DAVID MACHADO

Name of Dairy Facility: DAVID MACHADO

Facility Address: 26230 S. UNION RD. MANTECA 95337
Number and Street City Zip Code

Contact Person Name and Phone Number: DAVID MACHADO 209-581-1713
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

RELM DAIRY 6721 E. PERRIN RD. MANTECA 95337 209-599-2565
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

257-013-010
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 3/25/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 3,024,000 Gallons

Method used to determine volume of process wastewater: _____

42 HOURS @ 1200 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

Hauler's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: DAVID MACHADO

Name of Dairy Facility: DAVID MACHADO

Facility Address: 26230 S. UNION RD. MANTECA 95337
Number and Street City Zip Code

Contact Person Name and Phone Number: DAVID MACHADO 209-581-1713
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

RELM DAIRY 6721 E. PERRIN RD. MANTECA 95337 209-599-2565
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

257-013-010
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 5/24/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 2,412,000 Gallons

Method used to determine volume of process wastewater: _____

33.5 HOURS @ 1200 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

Hauler's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: DAVID MACHADO

Name of Dairy Facility: DAVID MACHADO

Facility Address: 26230 S. UNION RD. MANTECA 95337
Number and Street City Zip Code

Contact Person Name and Phone Number: DAVID MACHADO 209-581-1713
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

RELM DAIRY 6721 E. PERRIN RD. MANTECA 95337 209-599-2565
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

257-013-010
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 7/17/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 2,304,000 Gallons

Method used to determine volume of process wastewater: _____

32 HOURS @ 1200 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

Hauler's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: DAVID MACHADO

Name of Dairy Facility: DAVID MACHADO

Facility Address: 26230 S. UNION RD. MANTECA 95337
Number and Street City Zip Code

Contact Person Name and Phone Number: DAVID MACHADO 209-581-1713
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

RELM DAIRY 6721 E. PERRIN RD. MANTECA 95337 209-599-2565
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

257-013-010
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 8/10/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 2,304,000 Gallons

Method used to determine volume of process wastewater: _____

32 HOURS @ 1200 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

Hauler's Signature: David A. Machado
David A. Machado (Apr 1, 2024 17:23 PDT)
Date: Apr 1, 2024

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Brasil/Machado (F & S Brasil)
6524 Perrin Rd
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Dairy #1
RECEIVED DATE: 3/3/23 3:35 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/6/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW	T3062003A	Electrical Conductivity (EC)	EPA 120.1 and	1.37	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	45.2	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	745	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	48.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	876	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	92.6			0.000 (T3062003A)	0.000 (null)	
Electrical Conductivity (EC)	101			2.44 (T3062002B)	2.44 (Dup)	0.041
Nitrate Nitrogen (NO3-N)	101	82.5 (T3065001A)	80.6 (T3065001A)			2.38
Total Dissolved Solids (TDS)	105			745 (T3062003A)	770 (Dup)	3.30
Total Nitrogen (TN)	99.7	104 (null)	101 (null)			3.28

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Brasil/Machado (F & S Brasil)
6524 Perrin Rd
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Dairy #1
RECEIVED DATE: 3/3/23 3:35 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/6/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Heifers	T3062003B	Electrical Conductivity (EC)	EPA 120.1 and	2.22	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	55.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1420	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	101			0.041
Nitrate Nitrogen (NO3-N)	101	82.5 (T3065001A)	80.6 (T3065001A)	2.38

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Brasil/Machado (F & S Brasil)
6524 Perrin Rd
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Dairy #1
RECEIVED DATE: 3/3/23 3:35 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/6/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Dairy	T3062003C	Electrical Conductivity (EC)	EPA 120.1 and	0.830	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	531	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	101			0.041
Nitrate Nitrogen (NO3-N)	101	82.5 (T3065001A)	80.6 (T3065001A)	2.38

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: F+S BRASIL
Mailing address:
(not facility address) MANTECA CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID: DAIRY # 1

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here If performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	3/3/23	9:40	IW	AG WELL	16.0°C						H7
2		9:45	DW- HEIFERS	DOMESTIC	16.8°C		X				
3		9:50	DW - DAIRY	"	16.8°C		X				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-3-23 3:35			T3062003

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

David Machado 2023 Annual Report

Final Audit Report 2024-04-02

Created:	2024-04-02
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA_4y1h9d1t4BCHOT_exHg2wYKE8CzIf58

"David Machado 2023 Annual Report" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-04-02 - 0:07:27 AM GMT

Document emailed to hswoc@aol.com for signature
2024-04-02 - 0:11:00 AM GMT

Email viewed by hswoc@aol.com
2024-04-02 - 0:21:45 AM GMT

Signer hswoc@aol.com entered name at signing as David A. Machado
2024-04-02 - 0:23:35 AM GMT

Document e-signed by David A. Machado (hswoc@aol.com)
Signature Date: 2024-04-02 - 0:23:37 AM GMT - Time Source: server

Agreement completed.
2024-04-02 - 0:23:37 AM GMT

Adobe Acrobat Sign